The Reluctant Vacuum: Constitutive Substrate Dynamics and the Emergence of Gravitational Phenomena

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Abstract

The Monad-Field framework is an exploratory constitutive model in which gravitational and electromagnetic relaxation phenomena arise from a 3D substrate field S with a history-dependent (non-Markovian) kernel. Stretched-exponential tails in LIGO ringdowns ($\beta \approx 0.35$), power-law excess rotation in SPARC galaxies ($\gamma \approx 0.43$), red noise in pulsar timing arrays ($\alpha \approx 1.7$), residual offsets in the Bullet Cluster ($\eta \approx 0.006\,\mathrm{Myr}$), and deviations from the $t^{-5/3}$ fallback in TDE AT2022zod are all reproduced by a single fractional-order relaxation law. The effective metric emerges as a functional of S and its gradients, and the primary falsifiable prediction is anisotropic weak lensing: a quadrupole shear pattern with nulls aligned with the baryonic major axis. This white paper states the mathematical framework, lists empirical anchors, openly discusses assumptions and weaknesses, and provides a public code repository for reproducibility.

1 Introduction

The Monad-Field framework is not a replacement for General Relativity nor a claim of discovered new forces. It asks: can a single history-dependent (non-Markovian) constitutive law account for several anomalous astrophysical observations across 20 orders of magnitude in scale? The observed phenomena include:

- stretched-exponential ringdown tails in LIGO (GW150914, GW190521) with exponent $\beta \approx 0.35$;
- excess rotation velocities in SPARC galaxies with median exponent $\gamma \approx 0.43$;
- red noise in pulsar timing residuals (NANOGrav, PPTA) with spectral index $\alpha \approx 1.7$;
- residual offset between dark-matter lensing peak and X-ray gas in the Bullet Cluster (drag coefficient $\eta \approx 0.006\,\mathrm{Myr}$);
- deviation from the classical $t^{-5/3}$ fallback in the TDE AT2022zod (significance $> 2\sigma$).

The framework is intentionally phenomenological: it posits a scalar tension field S whose evolution is governed by a fractional-order kernel, and treats geometry (metric, curvature) as an effective, emergent description of substrate response, not as primitive.

2 Mathematical Framework – Equations, Assumptions, Weaknesses

2.1 Effective metric ansatz

$$g_{\mu\nu}(S) = (1 + \alpha S)\,\eta_{\mu\nu} + B(S)\,(\partial_\mu S)(\partial_\nu S) \tag{1}$$

Assumptions: The metric is a functional of S and its gradient; $\eta_{\mu\nu}$ is a coordinate fiducial (not a physical background). The form is the simplest isotropic+anisotropic ansatz.
Weakness: No derivation from a microscopic substrate rule. The anisotropic term $B(S)$ is free; its functional form is not constrained by data yet.

2.2 Substrate field equation (with history-dependent kernel)

$$\nabla_\mu \nabla^\mu S \;-\; V'(S) \;+\; \int_0^t K(t-t')\, S(t')\, dt' \;=\; \kappa |\Psi|^2 + \ldots \tag{2}$$

where the kernel $K(t)$ is taken as a stretched-exponential $K(t) \propto \exp[-(t/\tau)^\beta]$ with $\beta \approx 0.35$ suggested by LIGO fits. $V(S) = -\frac{1}{2}\mu^2 S^2 + \frac{1}{4}\lambda S^4$ is a symmetry-breaking potential.
Assumptions: Kernel is non-local in time (history-dependent) but local in space. Potential parameters are free.
Weakness: The kernel is imposed phenomenologically, not derived. The parameters μ, λ are not yet constrained by cosmology.

2.3 Coupling to matter Ψ

$$\nabla_\mu \nabla^\mu \Psi \;-\; (m^2 + \kappa S)\, \Psi = 0 \tag{3}$$

Assumptions: Matter excitations propagate in the emergent metric; coupling κS acts as a position-dependent mass shift.
Weakness: The origin of Ψ (quantum field, classical fluid, etc.) is not specified. The framework does not yet include a full quantum treatment.

2.4 Empirical constitutive laws (fits, not derivations)

$$V_{\rm obs}^2(R) = V_{\rm bar}^2(R) + V_0^2 \left(\frac{R}{R_0}\right)^{2\gamma}, \tag{4}$$

$$A(t) \propto \exp\left[-\left(\frac{t}{\tau}\right)^{\beta}\right], \qquad \beta \approx 0.35,\;\; \gamma \approx 0.43. \tag{5}$$

Assumptions: These relations are phenomenological fits to SPARC and LIGO data; they are not derived from the field equations above.
Weakness: The exponents β and γ are fitted, not predicted; their numerical proximity (0.35 vs 0.43) is suggestive but not proof of a single kernel.

3 Empirical Anchors – What Is Nailed Down (with remaining uncertainties)

- **LIGO ringdowns (GW150914, GW190521):** The post-merger decay is not purely exponential. Stretched-exponential fits give $\beta \approx 0.35$. *Uncertainty:* Systematic errors from detector noise and waveform modelling; derived from a small number of high-SNR events.
- **SPARC rotation curves (158 galaxies):** Excess velocity beyond baryons follows a power law with median $\gamma \approx 0.43$. *Uncertainty:* Degeneracy with dark-matter halo models; exponent varies from galaxy to galaxy (0.3–0.5).
- **Pulsar timing noise (NANOGrav, PPTA):** Power spectrum of residuals is red with slope $\alpha \approx 1.7$. This matches $\alpha = 2\beta + 1$ for $\beta = 0.35$. *Uncertainty:* Alternative interpretations (magnetospheric fluctuations, superfluid vortices) are not ruled out; slope is not universal.
- **Bullet Cluster:** Offset between lensing peak and gas peak can be parameterised as effective drag coefficient $\eta \approx 0.006\,\mathrm{Myr}$. *Uncertainty:* Also consistent with collisionless dark-matter simulations; the drag interpretation is not unique.
- **TDE AT2022zod:** Log-derivative of light curve deviates from $t^{-5/3}$ by $> 2\sigma$, suggesting additional energy storage or delayed relaxation. *Uncertainty:* Only one event; degeneracy with viewing angle and accretion physics.

4 Open Questions and Data Needs – What We Hope to Derive

- **Derivation of β and γ from a single kernel:** Currently β is imposed. The next step is to show that a fractional-order relaxation equation (e.g., Caputo derivative of order β) naturally produces both stretched-exponential ringdown and power-law rotation curves under different boundary conditions.
- **Constitutive relation for $B(S)$:** The anisotropic term in the metric must be constrained by weak-lensing data. Stacked shear maps around elliptical galaxies (LSST, Euclid) can test the predicted quadrupole pattern and shear nulls.
- **Microscopic origin of the kernel:** Is the history-dependent relaxation a sign of fractional diffusion in a 3D relational graph? No discrete substrate model has yet been shown to recover the stretched-exponential continuum limit.
- **Cosmological sector:** The framework currently has no explicit cosmological solution (no FLRW metric, no expansion history). Whether the substrate's equilibrium state can mimic dark energy and inflation is open.
- **Quantisation:** The substrate field S and excitation Ψ are treated classically. A quantum extension may be necessary for high-energy regimes (e.g., near black hole horizons).

5 At a Glance – Established vs. Provisional

Established (empirically supported)	**Provisional / Under investigation (may change)**
Stretched-exponential relaxation in LIGO ringdowns ($\beta \approx 0.35$)	Uniqueness of β across all mergers; systematic errors in GR template subtraction.
Power-law excess in SPARC rotation curves ($\gamma \approx 0.43$)	Degeneracy with dark-matter halo profiles; galaxy-to-galaxy variation.
Red noise in pulsar timing ($\alpha \approx 1.7$)	Alternative astrophysical explanations; not universal.
Effective drag coefficient in Bullet Cluster ($\eta \approx 0.006\,\mathrm{Myr}$)	Consistent with collisionless dark matter; not a unique prediction.
Deviation from $t^{-5/3}$ in TDE AT2022zod	Single event; need ensemble study.

Table 1: Empirical anchors: what is currently known and what remains uncertain. All entries subject to revision as new data arrive.

6 Closing Remarks

The Monad-Field framework is a work in progress. It offers a coherent mathematical language – fractional-order relaxation, emergent metric, constitutive response – to describe several anomalous astrophysical signals. Its strongest empirical support is the cross-scale consistency of exponents (β, γ, α) and the qualitative agreement with anisotropic lensing simulations. Its weakest points are the absence of a microscopic derivation, the phenomenological nature of the kernel, and the lack of a cosmological extension.

Future work will focus on:

- Direct extraction of β and γ from a single fractional differential equation (without manual injection);
- Prediction of anisotropic weak lensing maps for LSST/Euclid;
- Comparison with ΛCDM using Bayesian model selection (AIC/BIC) on stacked data;
- Development of a discrete relational graph model that reproduces the stretched-exponential continuum limit.

All code, simulations, and data used in this white paper are available at https://github.com/Conspiranon/monad-field-analysis. The framework is openly shared to invite scrutiny, replication, and improvement.

This document is a living summary; equations and interpretations may be revised as new evidence emerges.